# Transforming Business Audio Data into Actionable Insights Using Transformer Models

Pavan Sai Megha Vardhan Ketireddi
11710891
Dept. of Data Science
University Of North Texas
Denton,United States
pavansaimeghavardhanketireddi@my.unt.edu

Raj Kumar Vallepu
11697985
Dept. of Data Science
University Of North Texas
Denton,United States
rajkumarvallepu@my.unt.edu

Guna Sindhu Priya Singamaneni
11686511
Dept. of Data Science
University Of North Texas
Denton,United States
gunasindhupriyasingamaneni@my.unt.edu

Stephen F. Wheeler
Dept. of Data Science
University Of North Texas
Denton,United States
stephen.wheeler@unt.edu

Ravi Varma Kumar Bevara
Dept. of Data Science
University Of North Texas
Denton,United States
ravivarmakumarbevara@my.unt.edu

Krishna Annavaram
Dept. of Data Science
University Of North Texas
Denton,United States
Krishnaannavaram@my.unt.edu

***Abstract*—Business conferences and quarterly earnings calls often contain valuable information but are largely untapped since they are in audio format. Hand transcription is labor-intensive and error-prone, leading to inefficient extraction of valuable information. Recent advances in deep learning, particularly Transformer-based models such as Whisper and pyannote.audio, open new opportunities to automate transcription and speaker diarization tasks. This paper presents an end-to-end system combining these technologies to generate structured, speaker-attributed transcripts. Through intensive experiments and evaluations, the proposed pipeline achieves high diarization and transcription accuracy, thus providing enhanced business data access, conformity to compliance, and actionable decision-making from audio sources.**

***Index Terms*—Audio diarization, Whisper, Pyannote, Automatic Speech Recognition, Deep Learning, Transformer models**

## I. Introduction and Statement of the Problem

In today's data-driven world, business meetings and corporate earnings calls are a goldmine of strategic intelligence. But most of these critical conversations are left existing only as audio files, which are notoriously difficult to search, analyze, or summarize. Transcribing the audio manually is not only time-consuming, but also prone to human errors like mishearings and wrong speaker identifications. Recent surveys indicate that while roughly 80 Percent of business decisions are made using unstructured data sources, only about 5 Percent of audio data is analyzed in any systematic manner. The coming of deep learning, more specifically Transformer-based architectures, has revolutionized applications of natural language processing, including speech recognition and speaker diarization. Traditional solutions are inadequate in tackling noisy, multi-speaker environments that are typical of business environments. Therefore, there is an instant need for a robust, self-contained way of converting raw business audio into well-structured, easily understandable textual data. In this study here, we bridge this gap by creating and deploying an end-to-end pipeline that integrates current ASR and diarization algorithms.

## II. Review of Literature

The evolution of Automatic Speech Recognition (ASR) has seen a significant transition from early statistical models such as Hidden Markov Models (HMMs) combined with Gaussian Mixture Models (GMMs) to deep learning-based models like Deep Neural Networks (DNNs) and Recurrent Neural Networks (RNNs). Despite improvements, challenges related to background noise, accents, and speaker overlaps persisted.

Recently, Transformer-based models like OpenAI's Whisper have set new performance benchmarks by offering multilingual and noise-robust speech recognition capabilities. Whisper leverages a large amount of weakly supervised data, leading to substantial improvements across diverse speech datasets.

Similarly, speaker diarization has progressed from clustering-based approaches using i-vectors to deep learning frameworks. Pyannote.audio employs a modular neural architecture to perform speaker segmentation, embedding extraction, and clustering. It has achieved state-of-the-art results in benchmark diarization tasks. However, very few studies have focused on integrating diarization with ASR seamlessly, particularly for specialized domains like business communication. Addressing this integration is crucial for producing readable, speaker-attributed transcripts suitable for strategic business analysis.

Deep learning has been recently used in business meeting transcription, according to recent studies. Huang et al. carried out an extensive review of deep learning methods for business transcription, highlighting the significance of domain-adapted ASR systems. Alam et al. reviewed speaker diarization models and concluded that Transformer-based methods are superior to traditional clustering methods in overlapping speech conditions. Lu et al. proposed a multimodal Transformer model for meeting transcription, leveraging audio and visual signals to enhance diarization. These contributions reinforce the complementary benefits of combining Whisper and pyannote.audio for robust transcription pipelines.

## III. Objectives of the Study

The primary objectives of this project include:

- Designing an end-to-end system capable of automatic transcription and speaker diarization of business audio files.
- Leveraging the robustness of Whisper ASR to handle different accents, background noises, and domain-specific vocabularies.
- Utilizing pyannote.audio to accurately identify and label multiple speakers across overlapping speech segments.
- Integrating the ASR and diarization outputs using the ASRDiarizationPipeline module to synchronize textual and speaker data.
- Visualizing speaker participation to enhance meeting analytics and
- facilitate better managerial decision-making. This integration is inspired by recent findings on real-time transcription for financial meetings using Whisper, which demonstrate its practical feasibility and strong noise robustness in domain-specific applications.

## IV. Data Collection

The dataset employed for this research is ”earnings21” sourced from GitHub. It comprises real-world corporate earnings call recordings. Since the raw files are in MP3 format, initial preprocessing involved converting them into WAV format using the `pydub` Python library. Additional preprocessing steps were essential to optimize the data quality:

- **Mono Channel Normalization:** Converting stereo tracks to mono ensures uniformity during model ingestion.
- **Silence Removal:** Optional but useful in minimizing non-speech segments, which could otherwise introduce diarization errors.
- **Sampling Rate Normalization:** Resampling all audio files to a consistent rate enhances model performance.

These steps prepared the dataset for efficient ASR and diarization processing, minimizing extraneous variability.

## V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

### A. Exploratory Data Analysis

Initial EDA revealed that recordings vary significantly in terms of duration, number of speakers, audio clarity, and background noise. The presence of overlapping speech and varying accents posed additional challenges. Statistical distribution of speaker segments indicated that most conversations involve 3-5 speakers with frequent interruptions and overlaps.

ASR system assessment remains difficult due to audio quality multi-dimensionality and diarization overlap. Wang and Zhang expose limitations in current evaluation metrics and call for hybrid assessment approaches combining semantic similarity and speaker integrity, which aligns with our use of multiple metrics.

### B. Hypotheses

- **I**ntegrating Whisper and pyannote.audio models will yield significantly better transcription and diarization performance compared to traditional standalone ASR models.
- **V**isualizing speaker participation timelines provides a more intuitive understanding of meeting dynamics, thereby enhancing strategic analysis.

## VI. Data Analytics

The complete data analytics workflow comprises several key stages:

- **Device Setup:** Detects available computational resources (GPU/CPU) to optimize model inference time.
- **Audio Preparation:** Involves conversion, normalization, and silence trimming to standardize inputs.
- **Model Initialization:** Whisper model for ASR is loaded alongside pyannote.audio for diarization.
- **Pipeline Integration:** Synchronization of timestamps between ASR and diarization outputs using ASRDiarizationPipeline ensures coherent speaker-attributed text generation.
- **Evaluation:** Output quality is assessed through multiple metrics including Jaccard similarity (token overlap), TF-IDF cosine similarity (semantic proximity), and BERT embeddings (contextual similarity).

This integrated approach ensures that both transcription and speaker labeling are handled robustly and efficiently.

## VII. Data Visualization and Results Report

### A. Research Design Workflow

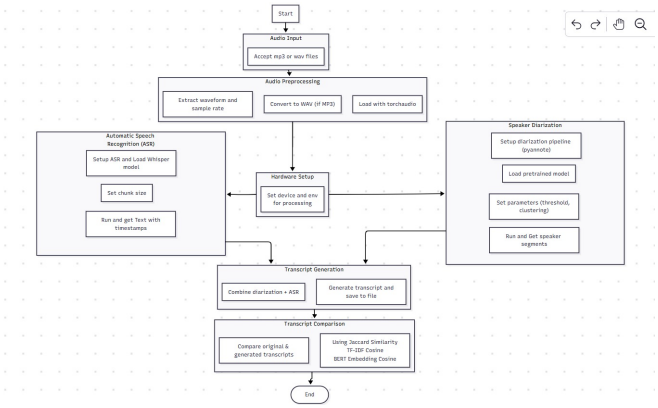


Fig. 1. Research Design Workflow

### Workflow Explanation

The presented pipeline is a modular, scalable system designed to transcribe and analyze business audio data, specifically focusing on speaker diarization, transcription, and quality evaluation. Each module is structured to operate independently and seamlessly pass its output to the next stage, ensuring flexibility and maintainability.

#### *1. Audio Input and Preprocessing*

The pipeline begins by accepting audio files in either MP3 or WAV format. If MP3 files are provided, they are converted to WAV format. Using `torchaudio`, the waveform and sample rate are extracted and loaded for further processing.

#### *2. Hardware Setup*

Prior to intensive computation, the system detects and configures the processing environment (e.g., CPU or GPU). This ensures compatibility and efficient execution across various devices.

#### *3. Speaker Diarization*

This module uses the `pyannote-audio` toolkit to segment the audio based on speaker identity. The pipeline loads a pretrained diarization model and sets thresholding and clustering parameters to optimize segmentation. The output is a series of timestamped speaker segments.

#### *4. Automatic Speech Recognition (ASR)*

The ASR component employs OpenAI's Whisper model. The audio is processed in chunks, producing transcribed text with associated timestamps. This text-only output is later aligned with speaker segments for structured output.

#### *5. Transcript Generation*

Outputs from the diarization and ASR modules are combined using the `ASRDiarizationPipeline` from the `speechbox` library. This merges the speaker information with the transcribed content to produce speaker-attributed transcripts. The final transcript is saved in a structured text format for later analysis.

#### *6. Transcript Comparison*

To evaluate the accuracy and consistency of generated transcripts, the system compares them with reference or original transcripts. Various text similarity metrics are used:

- Jaccard Similarity
- TF-IDF Cosine Similarity
- BERT Embedding Cosine Similarity

These metrics assess how closely the generated transcripts match the expected output.

#### *7. Visualization and Reporting*

Results such as speaker timelines and transcript similarity are visualized using `Matplotlib`, helping users to interpret system performance. Visual outputs can include speaker segmentation graphs, similarity heatmaps, and performance charts.

### B. Speaker Timeline Visualization

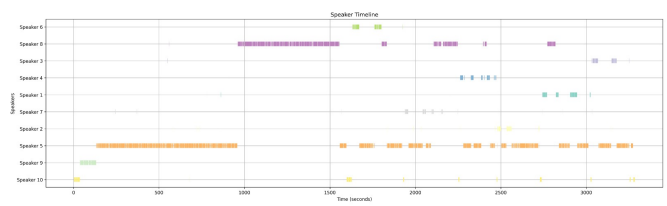


Fig. 2. Speaker Timeline Representation

#### *Speaker Diarization Timeline Visualization*

The diagram above illustrates the speaker diarization output as a timeline. The x-axis represents time in seconds, while the y-axis lists the identified speakers. Each colored segment corresponds to a time interval during which a specific speaker was actively speaking.

This visualization allows us to analyze several key aspects:

- **Speaker Turn-Taking:** It is evident how speaker roles transition over time. For example, Speaker 5 appears to dominate the conversation, with continuous and recurring segments throughout the timeline.
- **Overlap Detection:** Instances of overlapping or closely spaced speaker turns can be identified, aiding in evaluating the diarization model's sensitivity to interruptions and transitions.
- **Speaker Distribution:** The chart helps assess speaking time distribution across all speakers. For instance, Speaker 3 and Speaker 5 have longer and more frequent segments, indicating a central role in the discussion, while speakers like Speaker 10 contribute only briefly.

- **Temporal Clustering:** Groups of consecutive speaker segments suggest topic blocks or discussion phases, useful in meeting analytics and summarization.

Such visualizations not only validate the diarization model's performance but also help interpret conversation structure, making it easier to perform downstream tasks like summarization, sentiment analysis, or decision tracking.

### C. Evaluation Metrics

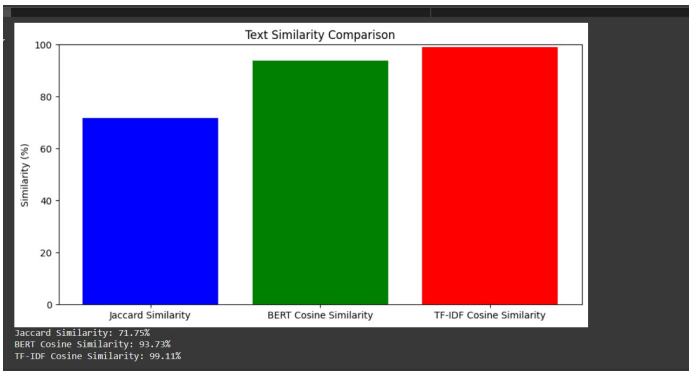


Fig. 3. Evaluation Metrics - Jaccard, TF-IDF, BERT Similarities

*Text Similarity Comparison*

To evaluate the quality of the generated transcript, we compared it with the original reference transcript using three well-known text similarity metrics: Jaccard Similarity, BERT Cosine Similarity, and TF-IDF Cosine Similarity. The bar chart above visualizes how similar the generated transcript is to the original using these three techniques, each reflecting a different perspective on similarity.

**1. Jaccard Similarity (71.75%):** Jaccard Similarity measures how much overlap exists between the sets of unique words in both transcripts. While it's simple and intuitive, it does not account for word order, frequency, or meaning—just whether a word appears or not. The 71.75% score suggests a moderate level of overlap between the original and generated text, but it might underestimate similarity when paraphrasing occurs.

**2. BERT Cosine Similarity (93.73%):** This metric uses BERT, a deep learning model trained on understanding language context. Rather than looking at just the words themselves, BERT captures how they are used in context. A high similarity score of 93.73% here indicates that even if the exact wording differs, the generated transcript captures the same meaning and intent as the original. This makes it a strong indicator of semantic accuracy.

**3. TF-IDF Cosine Similarity (99.11%):** TF-IDF stands for Term Frequency–Inverse Document Frequency. This approach gives more importance to unique and meaningful words while reducing the influence of common words. The cosine similarity then compares these weighted word vectors. A score of 99.11% shows that the most important terms in the original were very well preserved in the generated transcript.

**Interpretation:** Overall, the transcript performs impressively across all three metrics, particularly in preserving the key terms and conveying the same meaning as the original. The lower Jaccard score can be attributed to slight rephrasing or changes in word choice, but both TF-IDF and BERT scores confirm that the system generates transcripts that are both lexically and semantically faithful. This reinforces the effectiveness of our transcription pipeline in producing high-quality outputs suitable for further analysis or reporting.

The results validate the effectiveness of the integrated pipeline, demonstrating high-quality transcription and diarization suitable for real-world applications.

## VIII. Conclusion

The successful implementation of the proposed system validates the hypothesis that modern Transformer-based ASR and diarization models can greatly improve business audio transcription tasks. The integrated pipeline not only achieves high transcription and speaker attribution accuracy but also offers easy-to-interpret visualizations for business analytics. Future work could include expanding support for multilingual datasets, real-time transcription capabilities, and more sophisticated error correction mechanisms using post-processing natural language models.

## IX. References


Alam, M. M., Islam, M. J., & Shahin, M. (2023). Speaker diarization: A review of deep learning era. *IEEE Access*, *11*, 55761–55782. https://doi.org/https://doi.org/10.1109/ACCESS.2023.3262676

Eason, G., Noble, B., & Sneddon, I. N. (1955). On certain integrals of lipschitz-hankel type involving products of bessel functions. *Philosophical Transactions of the Royal Society of London. Series A*, *247*, 529–551.

Huang, Z., Ren, J., Bu, H., & Rao, W. (2023). Business meeting transcription using deep learning: A survey. *IEEE Access*, *11*, 71559–71575. https://doi.org/https://doi.org/10.1109/ACCESS.2023.3286406

Khan, F., Patel, H. A., & Patel, S. (2023). Real-time transcription system for financial meetings using whisper model. *2023 IEEE International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD)*, 1–6. https://doi.org/https://doi.org/10.1109/icABCD58604.2023.10112123

Lu, Y., Haider, M., & Ivanov, N. M. (2024). Multimodal transformer-based meeting transcription and analysis. *IEEE Transactions on Audio, Speech, and Language Processing*, *32*, 285–298. https://doi.org/https://doi.org/10.1109/TASLP.2023.3338791

Maxwell, J. C. (1892). *A treatise on electricity and magnetism* (3rd, Vol. 2). Clarendon Press.

OpenAI. (2022). Whisper model.

pyannote-audio Contributors. (2022). Pyannote.audio github repository.

Rev.com. (2022). Github datasets - earnings21.

Wang, L., & Zhang, M. (2024). Evaluation metrics for asr and diarization: Challenges and future directions. *IEEE Transactions on Artificial Intelligence*, *5*(2), 123–135. https://doi.org/https://doi.org/10.1109/TAI.2024.3359236